Make-up Online on Basic + Advanced SQL

Time: 60 + 5 mins Marks: 6 * 5 = 30

*There are **6 questions** in total. Read all the questions carefully. A row from the desired output has been attached to every question for your better understanding of the display format.* ***This row may or may not be the top row of your answer.***

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1. Find employees in Europe who have a commission percentage greater than 0.2 and work in departments with more than 3 employees. Show employee_id, full_name, commission_pct, and department_name in descending order of commission_pct and ascending order of employee_id.

EMPLOYEE_ID	FULL_NAME	COMMISSION_PCT	DEPARTMENT_NAME
145	John Singh	0.4	Sales

2. For each job title, show the job_title, employee_id, and hire_date of the most recently hired employee(s). If a job has no employees, do not include it. Display in descending order of hire_date. If hire dates are the same, prioritize by job_title alphabetically, then by lower employee_id.

JOB_TITLE	EMPLOYEE_ID	HIRE_DATE
Sales Representative	167	4/21/2018, 12:00:00 AM

3. Find all employees whose salary is greater than the average salary of employees in at least three different job titles within the same city. Display employee_id, full_name, job_title, city, and salary in descending order of salary and ascending order of employee_id.

EMPLOYEE_ID	FULL_NAME	JOB_TITLE	CITY	SALARY
100	Steven King	President	Seattle	24000

4. For each city, calculate the average salary of employees working there. Categorize this average as 'Low' (avg salary ≤ 5000), 'Medium' (avg salary > 5000 and ≤ 8000), 'High' (avg salary > 8000 and ≤ 12000), or 'Very High' (avg salary > 12000). Display city, average_salary, and salary_category in descending order of average salary and ascending order of city. Include cities with no employees (avg should be 0, categorized as 'Low').

CITY	AVERAGE_SALARY	SALARY_CATEGORY
Munich	10000	High

5. For each department, show the department_id, department_name, count of employees hired in January, February, or March, and the average salary of those employees. Display only departments where the average salary is a whole number (integer) and exclude any departments with null values. Sort by count of employees ascending, then by average salary descending.

DEPARTMENT_ID	DEPARTMENT_NAME	EMPLOYEES_HIRED_Q1	AVG_SALARY
90	Executive	1	17000

6. For each job_id, display the job_id, count of employees in that job, minimum salary, maximum salary, salary range (max - min), and percentage difference between max and min salaries. Calculate percentage difference as ((max_salary - min_salary) / min_salary * 100) rounded to 2 decimal places. Show only jobs where the salary range is greater than 5000 and sort by percentage difference descending, then by job_id ascending.

JOB_ID	EMPLOYEE_COUNT	MIN_SALARY	MAX_SALARY	SALARY_RANGE	PERCENTAGE_DIFFERENCE
SA_REP	30	6100	11500	5400	88.52